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Photoelectron Spectroscopy (XPS) and Thermogravimetry (TG) of Pure and Supported Rhenium Oxides

1. Pure Rhenium Compounds

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Abstract. To investigate the interaction of rhenium with several supports, a preliminary study on pure rhenium compounds has been carried out in order to achieve a better understanding of their surface chemical properties.

To this end X-ray photoelectron spectroscopy (XPS) and thermogravimetry (TG) methods have been applied.

The results show that metallic rhenium, ReO_2 and ReO_3 are covered with Re(VII). By heating in H_2, NH_4ReO_4 and rhenium oxides are reduced to metallic rhenium in different temperature ranges. In the case of Re_2O_7 a spillover effect has been found when platinum is present. For ReO_3 and ReO_2 a morphological model has been described on the basis of combined XPS, TG and surface area measurements.

The binding energy values of the different rhenium oxidation states have been assessed and their variation with the oxidation number briefly discussed.

Photoelektronenspektroskopie (XPS) und Thermogravimetrie (TG) an reinen und auf Trägern aufgetragenen Rhenium Oxiden. 1. Reine Rhenium Verbindungen

Inhaltsübersicht. Zur Untersuchung der Wechselwirkung von Rhenium mit verschiedenen Trägern wird zunächst eine Untersuchung an reinen Rhenium-Verbindungen zum besseren Verständnis ihrer oberflächenchemischen Eigenschaften durchgeführt. Dafür wurde die Röntgenphotoelektronenspektroskopie (XPS) und Thermogravimetrie angewandt.

Die Ergebnisse zeigen, daß metallisches Rhenium, ReO_2 und ReO_3 mit Re^{VII} bedeckt sind. Durch erhitzen in H_2 werden NH_4ReO_4 und die Rheniumoxide in unterschiedlichen Temperaturbereichen zu metallischem Rhenium reduziert. Im Fall des Re_2O_7 wurde in Anwesenheit von Platin ein Spillovereffekt gefunden. Für ReO_3 und ReO_2 wird ein morphologisches Modell auf Grund der kombinierten XPS-, TG- und Oberflächengrößen-Messungen beschrieben.

Werte für die Bindungsenergien der verschiedenen Rheniumoxydationsstufen werden abgeschätzt und ihre Veränderung mit der Oxydationszahl kurz diskutiert.

Introduction

Rhenium containing catalysts have recently acquired great importance in many industrial applications, primarily in reforming processes mixed with noble metals, as well as for other reactions, such as metathesis and hydrogen production

[1—5]. The metal is used in a dispersed form over a carrier. It is well known that the carrier does not act simply a dispersion [6, 7]: in fact the dispersion capability often arises from an interaction with the support. The carrier also influences the oxidation-reduction processes of the metal and of its oxides, and its oxidation state is therefore determined not only by the prevailing conditions but also by the nature of the support [8, 9].

The interaction of rhenium with the support has been the object of several investigations [10—19]. In a study of supported rhenium catalysts we have examined a number of supports, TiO_2, MgO, SiO_2, Al_2O_3 and a preliminary note on some of the thermogravimetric (TG) results has been published [20]. The study has been extended to include X-ray photoelectron spectroscopy measurements (XPS). The XPS and TG techniques are largely complementary, and it was a point of interest to investigate the potential of a combined study. In fact, TG allows accurate measurement of the oxygen loss (or acquisition) processes, but it does not easily give information on the different stages occurring on the surface, by which the process is accomplished. On the other hand, XPS provides information on the state of the outer layers, and hence on the surface processes, failing however in most cases to allow a quantitative description of the stoichiometry to be made.

The study of supported rhenium requires a comparison with the behaviour of pure oxides. The chemistry of rhenium is fairly complex, and it is only in recent years that a better understanding has been achieved [21, 22]. Literature data on the XPS of rhenium oxides are scanty [23—25] and it is not possible to obtain a clear picture of the binding energies (BE) of the different oxidation states. It was therefore necessary to first monitor the pure oxides and the results are presented and discussed in this paper. Subsequent papers deal with rhenium-containing specimens on various supports: TiO_2, MgO, SiO_2, Al_2O_3.

Experimental

Samples. The following commercial chemicals were used: NH_4ReO_4 (Ventron), Re_2O_7 (Merck), ReO_3 (Ventron), metallic rhenium foil (Ventron). ReO_2 was prepared by heating ammonium perrhenate at 520 °C for 3 h in a stream of nitrogen dried by passage through a liquid nitrogen trap and purified by passing over copper turnings at 400 °C [26]. Metallic rhenium powder was obtained by reduction of ammonium perrhenate in a hydrogen stream at 500 °C for 2 h and then at 1000 °C for further 2 h [27]. The following surface area values (Kr adsorption) were determined ($m^2 g^{-1}$): ReO_3, 1.1; ReO_2, 2.6; Re (powder), 1.0.

Photoelectron spectroscopy (XPS) measurements. XPS measurements were performed on AEI ES 100 and Vacuum Generator VG 3 spectrometers. The Al (1486.6 eV) excitation was used in both cases. The powdered specimen was either pressed into a through (6 · 15 · 2 mm) (AEI spectrometer) or it was spread on a gold-plated metallic sample holder (VG spectrometer). The C(1s) line (BE = 285 eV) was employed as reference. Excellent agreement was found on this basis (within 0.2 eV) between data collected with the AEI and VG instruments. In some experiments the samples were heated in a hydrogen atmosphere directly in the VG 3 spectrometer (p_{H_2} = 0.5 torr, temperature up to 420 °C, for 3 h, static conditions renewing the hydrogen charge every 15 minutes).

Thermogravimetric measurements. The thermogravimetric investigation was carried out by means of a Cahn RG electrobalance. The measurements were performed in a hydrogen stream

(20 cm^3/min) raising the temperature linearly, (3°/min) by a temperature programmer, up to 1000 °C. The gas was dried by a liquid nitrogen trap. In order to eliminate gas effects, caused by the asymmetric geometry of the system, the position of the zero was controlled in the whole temperature range by using a platinum dummy sample with a weight comparable to that of the studied samples. The measurements were performed by using either a platinum crucible or a silica container. The weight was measured to an accuracy of ±0.01 mg; and thus the weight changes to an accuracy of ±0.02 mg.

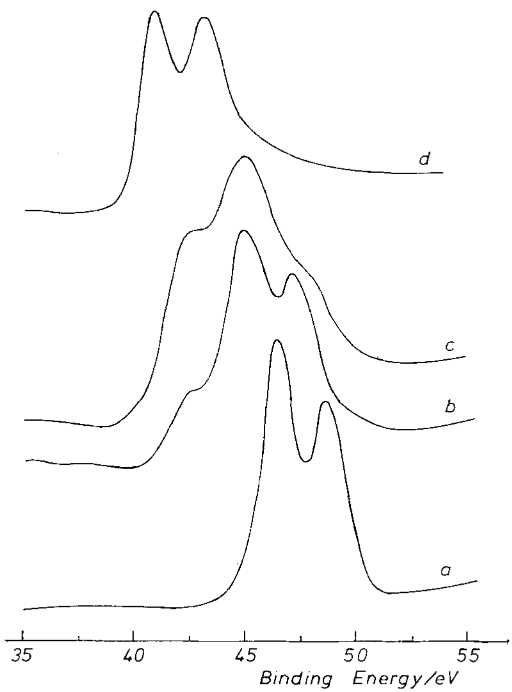
Results

A. XPS Studies

There can be some uncertainty in the determination of the BE values of rhenium in its different valency states not only due to difficulties inherent to the XPS technique, but also to the lack of definition of the state of the surface in commercial or in laboratory prepared specimens. It is therefore of interest to follow the evolution during reduction and/or oxidation thus allowing more stringent assignment of the species formed.

1. Ammonium perrhenate

The study of NH_4ReO_4 is a prerequisites in view of the fact that catalysts are usually prepared, as in the present case, by reduction of materials impregnated with ammonium perrhenate. The initial spectrum is shown in Fig. 1a.



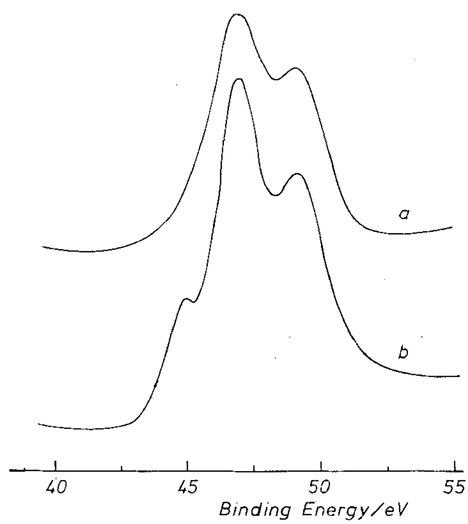


Fig. 1 Spectrum of NH_4ReO_4; a) initial spectrum; b) reduced with H_2 in situ (VG spectrometer) 250 °C, 1 h, P_{H_2} = 0.5 torr; c) reduced (as above) at 420 °C, 1 h; d) reduced 420 °C, 3 h; 0.5 torr

Fig. 2 Spectrum of Re_2O_7; a) initial spectrum; b) heated at 200 °C inside the spectrometer (AEI) for 15 min

Reduction of pure NH_4ReO_4 by hydrogen at 500 °C, which is the procedure adopted for supported catalysts, leads to metallic rhenium, as is shown by the TG technique (see below); it was not possible to adopt the same conditions as for the TG experiments inside the spectrometer. The reduction inside the VG spectrometer, carried out as described in the Experimental section, at 250 °C or 420 °C, showed that reduction did occur. The lowest BE peak $Re(4f_{7/2})$ was found at 40.7 eV, the presence of oxidized rhenium Re(VII), Re(VI) and Re(IV) was however apparent (curves b, c) and still visible as a tail in the final curve (d). This result appears surprising, since according to the TG data, ammonium perrhenate is reduced at 300 °C. The cause may be the less efficient reducing conditions, but the result also points to the ease of oxygen retention on the surface of the metal.

2. Re_2O_7

This oxide shows the $Re(4f_{7/2})$ peak at 46.9 eV (Fig. 2 curve a). A heating treatment inside the spectrometer shows the development first of a shoulder, then of a peak at 44.7 eV, shown to be due to ReO_3 (curve b).

3. ReO_3

The spectrum of this compound is composite (Fig. 3a) with two doublets of comparable intensity. The $Re(4f_{7/2})$ is at 44.5 eV for one component, whilst the second component is at 46.8 eV. A deconvolution of the composite peak shows that the intensity of each component is about 50%. A marked surface oxidation to Re_2O_7 is thus apparent. The specimen however is not simply oxidized in the surface. As TG experiment and X-ray analysis show (see below) the sample contains a large proportion of metallic phase which is not detectable by XPS, indicating that the metallic phase is fully covered by the oxide. Fig. 3 (curve b) shows the effect of rinsing the sample with water. A peak at 40.8 eV is now visible, amounting to about 7% of the total intensity, while on the high-BE side of the spectrum the signal ascribed to Re(VII) has decreased to about 27% of the total intensity.

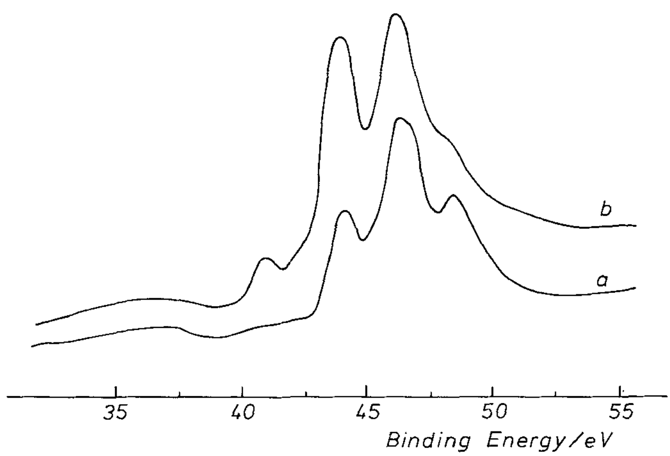


Fig. 3 Spectrum of ReO_3; a) initial spectrum; b) spectrum of ReO_3 rinsed with H_2O

4. ReO_2

This oxide also shows a composite pattern, one doublet having the $Re(4f_{7/2})$ peak at 42.5 eV plus a region which cannot be easily deconvoluted. This region may arise from two (or perhaps more) components in the BE region of the higher oxidation states, whose presence may be due either to incomplete reduction of the starting perrhenate, or to re-oxidation of ReO_2. Spectra recorded at different times after the preparation show that a slow re-oxidation of ReO_2 did in fact occur. A deconvolution has been attempted on the basis of the position of the peaks $4f_{7/2}$ for Re(VI) and for Re(VII), and a reasonable agreement has been found with the experimental curve by assigning to the components the relative intensities 72% (IV), 20% (VI), and 8% (VII). Reduction by hydrogen inside the spectrometer results in the spectrum of rhenium metal, with $Re(4f_{7/2})$ at 40.7 eV, as shown in Fig. 4. Curve a is the initial ReO_2 spectrum, referring to an oxide specimen first exposed to air for several months.

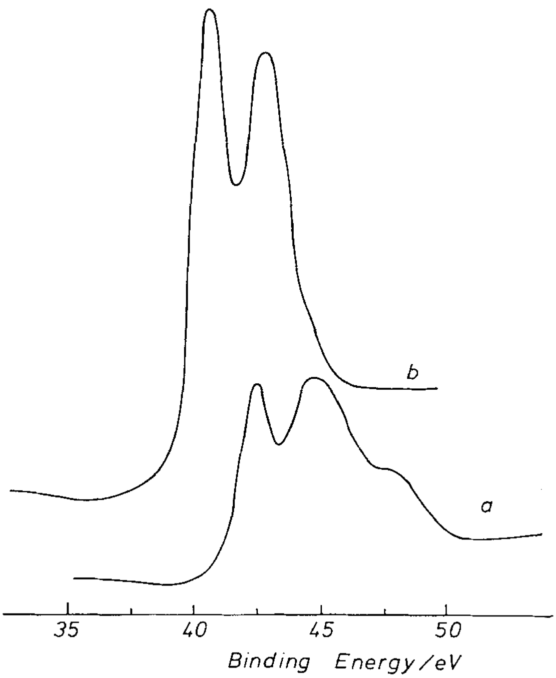


Fig. 4 Spectrum of ReO_2; a) initial spectrum; b) after reduction inside the spectrometer (VG) 420°C, 3 h, P_{H_2} = 0.5 torr

5. Re (metal)

A rhenium metal strip shows the presence of oxides including Re(VII) oxide, in addition to the metal, which is characterized by the $Re(4f_{7/2})$ peak at 40.7 eV (Fig. 5, curve a). Reduction by hydrogen at 500°C immediately before placing the specimen in the AEI spectrometer results in the spectrum of Fig. 5, curve b, which shows a sharp doublet, while the oxidized species have disappeared.

A rhenium metal powder shows a higher degree of oxidation (Fig. 6, curve a). Reduction inside the VG spectrometer, at 420°C, gave the spectrum shown in Fig. 6, curve b identical to that of Fig. 5, curve b.

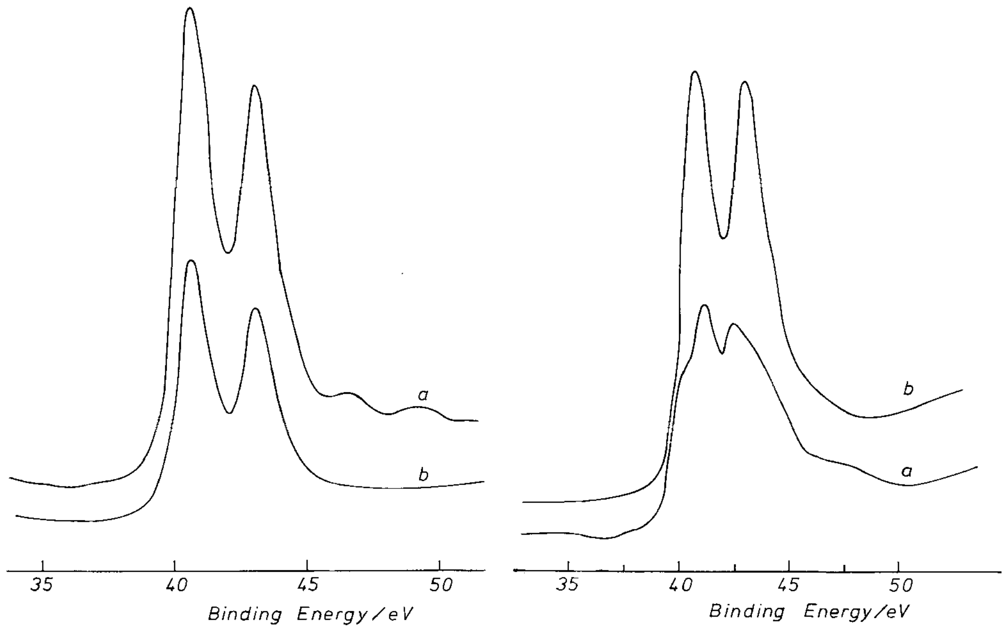


Fig. 5 Spectrum of Re metal (foil); a) initial spectrum; b) after reduction outside the spectrometer (AEI) with flowing H_2, atmospheric pressure at 500 °C, and immediate insertion in the analyzing chamber

Fig. 6 Spectrum of Re metal (powder); a) initial spectrum; b) reduced by H_2 in situ (VG) spectrometer; 420 °C, 3 h, $P_{H_2} = 0.5$ torr

B. Thermogravimetric Reduction and Oxidation Measurements

1. NH_4ReO_4 (reduction)

Ammonium perrhenate when heated in hydrogen is reported to give NH_3 and metallic rhenium [28] according to $2NH_4ReO_4 + 7H_2 \rightarrow 2NH_3 + 2Re + 8H_2O$. The reduction starts at 225 °C and is complete at 350 °C using either platinum or silica crucibles (Fig. 7, curve a). Table 1 shows the experimental and the calculated loss in weight. The experimental weight losses do not depend on the nature of the crucible used and are significantly lower (0.85%) than the calculated one. This result indicates that metallic rhenium tends to retain small amounts of oxygen.

Table 1 Thermogravimetric data for NH_4ReO_4, Re_2O_7 and ReO_2

Sample	Δ %calc.[a])	Temperature range of the weight loss	Δ %exp. (Pt)[a])	Δ %exp. (SiO_2)
NH_4ReO_4	30.57	225–350	30.31	30.32
Re_2O_7	23.12	120–270	23.22	44.33
ReO_2	14.66	120–425[b])	15.10	15.30

[a]) Loss in weight are expressed as % by weight. Δ% (Pt) and Δ% (SiO_2) weight losses obtained by using a platinum or a silica crucible, respectively.
[b]) In the temperature range 120–250 °C a slight weight loss (0.4%) is detected; see inset of Fig. 7.

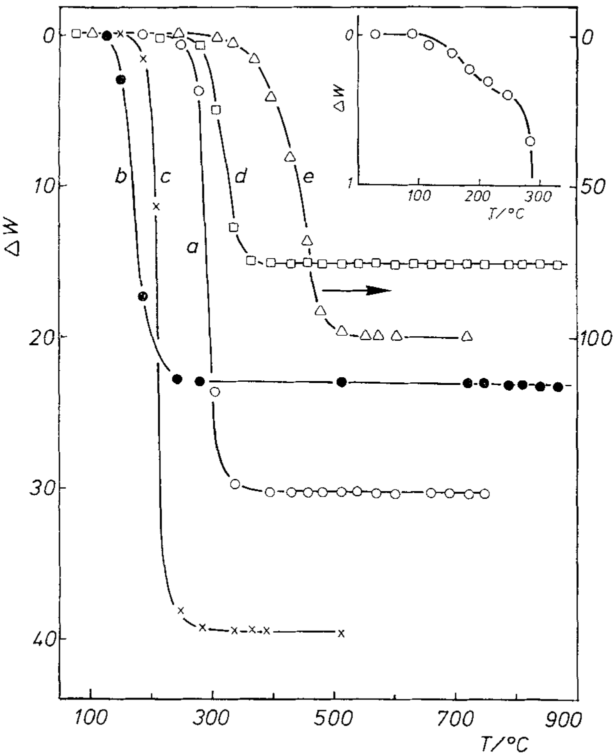


Fig. 7 Thermogravimetric weight loss, ΔW (mg/100 mg of sample) vs. T °C. ○ NH_4ReO_4; □ ReO_2; ● Re_2O_7 (Pt-crucible); × Re_2O_7 (silica crucible): left ordinate. △ Re (powder): right ordinate. The inset shows the details of the percentage weight loss, ΔW, for ReO_2 in the initial stage

2. Re_2O_7 (reduction)

As the heptaoxide is highly hygroscopic [29], the thermogravimetric measurement was performed after drying the symple in the thermogravimetric apparatus. To this end an oxygen stream (dried through a liquid oxygen trap), was allowed to flow in the thermogravimetric equipment and the temperature was raised slowly up to 120 °C. This temperature was kept constant until no weight change was observed. After evacuation, flowing hydrogen (dried through a liquid nitrogen trap) was admitted into the apparatus and the thermigravimetric measurement carried out. The results are different for the two types of crucible (Fig. 7, curves b, c).

By using the silica container a loss in weight higher than the calculated value for reduction to metallic rhenium was observed (Table 1). In addition, on the reactor wall a thin film having a metallic lustre was observed, as already noticed by Ratner et al. [28] during reduction of NH_4ReO_4.

With the platinum crucible, the weight loss starts around 120 °C and is complete at 250 °C. The observed value (23.30%) is close to that calculated (23.12%) on the basis of the rduction of the heptaoxide to metallic rhenium, Table 1. The slight discrepancy (0.18%) in excess may be accounted for by incomplete dryness of the starting heptaoxide.

It may be noted that in the literature [21, 29], on the basis of the work by Noddack [30], the heptaoxide is sometimes reported to be reduced by hydrogen to the dioxide at 300 °C and to the metal at 800 °C. It is clear that such a high temperature for the reduction to the metal, which disagrees with the results of the present work and with recent literature data [19, 31—33], is not correct.

3. ReO_3 (reduction)

From X-ray analysis the sample resulted to be a mixture of Re and ReO_3. From the width of the X-ray diffraction lines one can infer that the metal is in the form of small particles. From the XPS spectrum the presence of Re_2O_7 could also be demonstrated. The reduction of ReO_3 in hydrogen occurs in the range 120—280 °C and gives metallic rhenium (X-ray identification). By thermogravimetric measurements in hydrogen, performed before (Re_2O_7 present) and after rinsing the sample with water (only Re and ReO_3 present), the amounts of Re_2O_7, ReO_3 and Re were evaluated to be: ReO_3 42.3%, Re 43.4%, Re_2O_7 14.3%.

4. ReO_2 (reduction)

For both platinum and silica crucibles the reduction of ReO_2 occurs in the range 225—370 °C. However with both crucibles a very small loss in weight (about 0.4%) is also present in the temperature range 120—225 °C, as shown in the inset of Fig. 7. In addition, a small difference in the loss in weight is observed by using the two different crucibles: a higher value is obtained in the case of the silica container. Experimental values are higher than the calculated ones for the reduction to metallic rhenium (Table 1). The slight difference can be accounted for by recally that higher oxidation states are in fact stable and present (see XPS results), and therefore the dioxide surface retains extra oxygen.

5. Oxidation of rhenium

Fig. 7 (curve e) shows the thermogravimetric curve of Re powder oxidation in air. The loss in weight, due to the volatilization of Re_2O_7 obtained from the metal oxidation, begins at about 300 °C and is complete at 500 °C.

Discussion

We first examine the XPS results for the rhenium oxides. The TG results and the deductions which can be made from a comparison of the two techniques will be thereafter illustrated.

XPS data and BE values of rhenium compounds

As noted in the Introduction, in view of the interest in supported rhenium catalysts, it is necessary to know the binding energy (BE) values of unsupported compounds (in particular, the oxides) having different oxidation numbers (ON). The only extensive collection on rhenium compounds available in the literature refers to complexes [24], which are notably different in the nature of the bonds from those in rhenium oxides. The published data do not allow a clear picture to be drawn of the BE values vs ON. There are two reasons for this: first, the effect of charging, which heavily depends on the type of sample, secondly the difference of the type of bonds.

Effects due to charging are most likely on nonconducting oxides, and hence on Re_2O_7, and more markedly on the compound NH_4ReO_4. As mentioned above, the former oxide tends to show the presence of lower (conducting) oxides on its surface. This fact, while being a complicating feature of the spectrum, is likely to help to ensure a more uniform electric field on the surface and a good electrical contact. In fact, a constancy of the BE differences, ΔBE, between the C(1s) and the O(1s) peaks has been observed, and both peaks, though exhibiting an asymmetrical shape arising from indistinct compounds (carbonaceous and carbonate-like deposits, oxidic oxygen and hydroxyls or water) are reproducible. On NH_4ReO_4 on the other hand it is sometimes possible to observe a duplicate of O and of C peaks, whose intensity and separation appears erratic, and which can be attributed to the existence of distinct zones with different charging due to the bad electrical contact. As a consequence, the BE of Re(VII) in ammonium perrhenate can be assessed with lower accuracy.

As for the second consideration, BE values can be reasonably compared within a series of compounds of the same type, and the oxides constitute such a class.

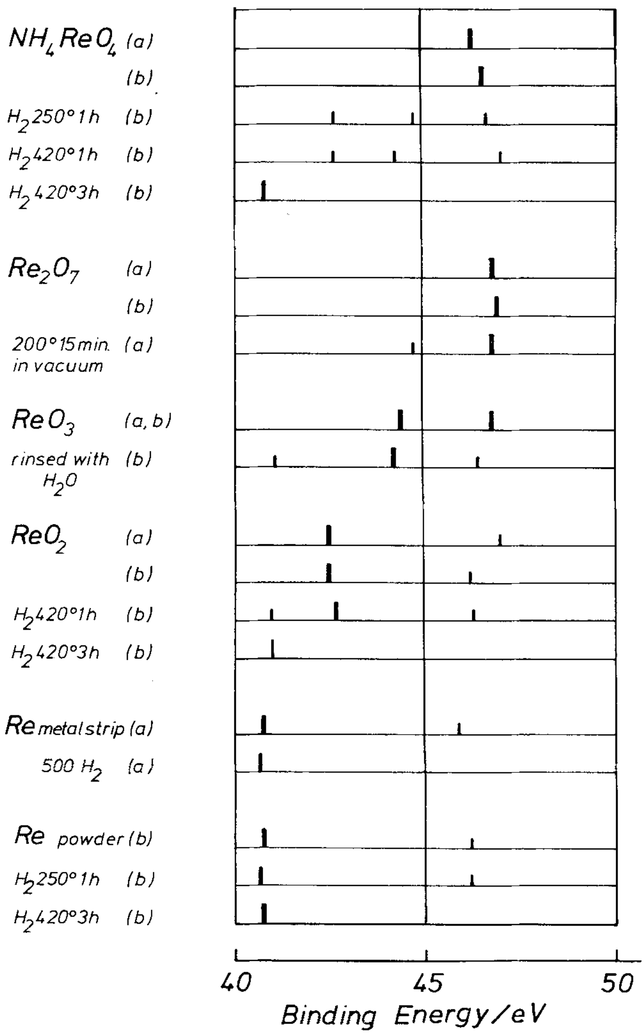


Fig. 8 A diagram of the B. E. values obtained from spectra recorded for the rhenium compounds: a) AEI spectrometer; b) VG spectrometer. Thick lines refer to the most intense peak position of Re $4f_{7/2}$, shorter and thinner lines refer to low intensity peaks and/or to unresolved shoulder (deconvoluted)

The results of Fig. 8, averaged for each compound regardless of the type of spectrometer, VG or AEI, give rise to the plot of Fig. 9. A lack of linear dependence of the BE on ON is apparent. A similar behaviour was observed for molybdenum oxides [34]. It can be observed that there is no a priori reason for linear dependence of BE on formal charge. A linearity between BE and effective charge can be predicted, if relaxation terms are constant [35]. Two observations can be made with regard to this question. First, the ON is a formal definition quite distinct from the actual charge of the ion, and accordingly a non-linear plot of BE vs ON may imply that the ON does not coincide with effective charge, as also suggested by the change in the nature of the metal-oxygen bond as ON increases. Second, the metallic properties of ReO_3 and ReO_2 should be considered, hence the highly polarizable conduction electrons cause a considerable relaxation in the final state of the XPS process, and the relaxation lowers the measured BE value. This explains the drop of BE when passing from Re_2O_7 (nonconducting) to ReO_3 or ReO_2.

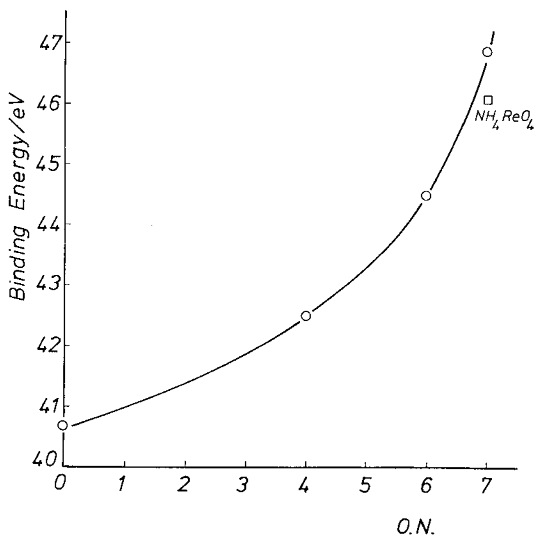


Fig. 9 A plot of B. E. for Re ($4f_{7/2}$) peak vs oxidation number ON

Thermogravimetric data and oxide stability

Thermogravimetric data provide a means for evaluating the relative proportions of the different rhenium oxides simultaneously present in the specimens. Together with the XPS data it is often possible to obtain a morphological picture of the oxides.

A complicating but interesting feature is offered by Re_2O_7, whose behaviour upon reduction depends on the nature of the crucible used. The higher loss in weight observed with the silica container (Table 1) must be ascribed to the partial volatilization of Re_2O_7. The volatilized oxide is then reduced in the vapour phase to lower oxides and to metal, as shown by the metallic lustre observed on the cold wall of the reactor in the TG apparatus. The high volatility of Re_2O_7, caused by its peculiar crystal structure, is known [36]. The observed behaviour indicates that the rate of reduction of the oxide may be slower than the rate of volatilization. The absence of this anomalous loss when platinum is present, shows that with the platinum crucible the reduction rate is increased. This finding points to a hydrogen spill-over effect. The spill-over effect in the reduction of Re_2O_7 has been reported [37]. If the volatilization of the heptaoxide is avoided (platinum crucible), the observed weight loss is that expected on the basis of the reaction $Re_2O_7 + 7\,H_2 = 2\,Re + 7\,H_2O$ within the error of likely and slight-water contamination. Despite the less efficient conditions for reduction inside the spectrometer, the influence of platinum on the reduction rate of Re_2O_7 is also evident from XPS measurements. The intensity ratios of Re(VI) and Re(VII) peaks ($4f_{7/2}$), after heating the Re_2O_7 sample in hydrogen at 100 °C, 0.31; 180°C, 1.25; 280 °C, 1.71; 320 °C, 1.9; spread on platinum sample holder or at 100 °C, 0.39; 180 °C, 0.79; 280 °C, 1.1; 320 °C, 2 on gold sample holder show that if platinum is present, in the temperature range 200–300 °C a higher degree of reduction is achieved. For temperatures higher than 300 °C, the reduction rate on gold becomes large enough to eliminate the influence of the sample holder.

For Re_2O_7 the weight loss is in excess (Table 1), therefore there is no indication that appreciable amounts of lower oxides are present. In this regard, recalling that the XPS data also show that there are no lower oxides on the surface of Re_2O_7, one is led to conclude that the oxidation tends to produce the heptaoxide as a stable outside phase.

The last conclusion is corroborated by results on the ReO_3 specimen, for which the presence of Re_2O_7 on the surface is clearly demonstrated by XPS spectra. Even after rinsing with water, which dissolves the heptaoxide, a thin layer of heptaoxide remains or is produced immediately, since the spectrum still shows the presence (though reduced in intensity) of the heptaoxide. It is also recalled that the commercial ReO_3 examined contains metallic rhenium, as shown by X-ray diffraction analysis and also by TG experiments. This fortuitous fact again provides an opportunity for demonstrating the complementarity between XPS and TG, in addition to confirming the stability of Re_2O_7. A key point is the appearance of the Re metal peaks in the XPS spectrum only after rinsing. From the TG data, and from the surface area (1.1 $m^2\,g^{-1}$) a uniformly sized spherical particle of radius $r = 2538$ Å is inferred. If a uniform thickness of Re_2O_7 is assumed, the layer would be 232 Å thick, which justifies the absence of the Re(0) peak in the XPS spectrum of ReO_3 before rinsing. Since the X-ray diffraction lines of Re(metal) are broad, small particles must be present. These may originate during the reduction of Re_2O_7, which leads to ReO_3. The morphology of ReO_3 specimen can be further specified thanks to the observed broadening of the Re metal X-ray diffraction lines. The estimate of the metal particle size can be made according to KLUG-ALEXANDER [38] assuming for simplicity the average between values arising from curves a and b therein quoted and taking into account the instrumental broadening measured on calibrated quartz particles. The estimates give values of metal particle diameters of about 400 ± 50 Å. The estimate of the ReO_3 according to the same method is affected by a much larger error, and gives a value of about 1200 ± 300 Å. The actual texture appears then to be large particles of ReO_3, embedding relatively smaller particles of Re metal with Re_2O_7 on the external of the agglomerate. There is no wonder that in such a "pudding", picture, some of the metal particle are closer to the surface, and once the Re_2O_7 layer is washed away they can show up at the XPS analysis. It is not our purpose however to dwell on this point, but merely to emphasize the complementarity of the techniques in the study of complex systems.

In the case of ReO_2, there is a small but significant weight loss (0.4%) in the temperature range 120—225 °C, which is the range of reduction of the heptaoxide see Table 1. This weight loss can therefore be attributed to reduction of Re_2O_7 present on the surface. In fact, by substracting this weight loss from the total variation observed at the end of the thermogram (15.10%) the value obtained (14.70%) is in excellent agreement with that calculated for the reduction of ReO_2 (14.66%). Furthermore, if the silica crucible is used instead of the platinum one, a larger loss in weight is observed (Table 1) since in the absence of spill-over phenomena the heptaoxide volatilizes instead of being reduced. The amount of Re_2O_7 thus calculated corresponds to 1.7% of the total specimen. By assuming a uniform size of spherical particles of ReO_2 with a radius, derived from the surface area (2.6 $m^2\,g^{-1}$) and density (11.4 $g\,cm^{-3}$), and a uniform heptaoxide layer, the thickness of the latter (density 7.4 $g\,cm^{-3}$) would be around 10 Å. It is now possible to avaluate the effect of this layer on the intensity of the underlying Re(IV). By assuming a constant mean free path λ of about 20 Å, a 10 Å thick layer reduces the intensity to 60% of its unreduced value. The intensity of the outer, oxidized layer is therefore 40%. As noted in the XPS section, the intensity of the Re(IV) peak was 72%, in agreement with the model. The XPS data, moreover, specify that the external layer actually contains both Re(VI) and Re(VII).

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